

Additional file 1: Supplementary tables

No Single Model Family Dominates: Ensembles and Conventional Machine Learning Perform Comparably to Pretrained Molecular Models Across 44 Property-Prediction Benchmarks

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This file contains the eleven supplementary tables referenced in the main text. Tables S1–S7 are generated directly from the deposited benchmark artifacts by `portable_colab_qsar_bundle/render_manuscript_assets.py` and `render_latex_tables.py`; Table S8 is generated by `python -m qsarena.reliability_tables` from the reliability-study artifacts; Tables S9–S11 are generated by `qsarena.meta_analysis`, which the same render script runs. They carry the same values as the machine-readable CSV and Markdown files in the repository.

All results derive from a single split and a single seed per dataset; as set out in the Limitations section of the main text, individual entries are therefore provisional and close comparisons between models should not be treated as established.

Table S1: Per-dataset winning model under the ranking metric, with the corresponding cross-validation-selected model and its relative gap to the dataset best (%).

dataset	suite	task_ kind	family	model	analysis_ metric	analysis_ metric_value	cv_ selected_ model	cv_ selected_ family	cv_selected_gap_to_best
tdc_ames	TDC	classification	TabPFN (tabular foundation)	TabPFNClassifier	test_ roc_ auc	0.876	TabPFNClassifier	TabPFN (tabular foundation)	0.000
tdc_bbb_martins	TDC	classification	Conventional ML	Random forest	test_ roc_ auc	0.925	TabPFNClassifier	TabPFN (tabular foundation)	1.707
tdc_ bioavailability_ ma	TDC	classification	Conventional ML	CatBoost	test_ roc_ auc	0.777	TabPFNClassifier	TabPFN (tabular foundation)	5.802
tdc_ carcinogens_ lagunin	TDC	classification	Chemprop v2 GNN	Chemprop v2 (AttentiveFP, ensemble=3)	test_ roc_ auc	0.929	TabPFNClassifier	TabPFN (tabular foundation)	6.339
tdc_clintox	TDC	classification	CFA combinatorial fusion	CFA (Combinatorial Fusion)	test_ roc_ auc	0.974	TabPFNClassifier	TabPFN (tabular foundation)	5.190
tdc_cyp1a2_ veith	TDC	classification	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5-fold))	test_ roc_ auc	0.973	ChemML MLP (Tensor Flow)	Deep tabular NN (ChemML MLP)	1.639
tdc_cyp2c19_ veith	TDC	classification	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5-fold))	test_ roc_ auc	0.931	XGBoost	Conventional ML	9.806
tdc_cyp2c9_ substrate_ carbonmangels	TDC	classification	Chemprop v2 GNN	Chemprop v2 (AttentiveFP, ensemble=3)	test_ auprc	0.438	SVC	Conventional ML	31.368
tdc_cyp2c9_ veith	TDC	classification	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5-fold))	test_ auprc	0.829	TabPFNClassifier	TabPFN (tabular foundation)	36.113
tdc_cyp2d6_ substrate_ carbonmangels	TDC	classification	Chemprop v2 GNN	Chemprop v2 (D-MPNN + RDKit2D, ensemble=3)	test_ auprc	0.693	AdaBoost	Conventional ML	10.520
tdc_cyp2d6_ veith	TDC	classification	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5-fold))	test_ auprc	0.761	TabPFNClassifier	TabPFN (tabular foundation)	39.882
tdc_cyp3a4_ substrate_ carbonmangels	TDC	classification	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5-fold))	test_ auprc	0.724	TabPFNClassifier	TabPFN (tabular foundation)	6.553
tdc_cyp3a4_ veith	TDC	classification	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5-fold))	test_ auprc	0.902	TabPFNClassifier	TabPFN (tabular foundation)	17.973

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dataset	suite	task_ kind	family	model	analysis_ metric	analysis_ metric_value	cv_ selected_ model	cv_ selected_ family	cv_selected_gap_to_best
tdc_dili	TDC	classification	Uni-Mol (3D pretrained)	Uni-Mol V1	test_ roc_ auc	0.922	Logistic Regression	Conventional ML	19.151
tdc_herg	TDC	classification	Chemprop v2 GNN	Chemprop v2 (AttentiveFP, ensemble=3)	test_ roc_ auc	0.857	TabPFNClassifier	TabPFN (tabular foundation)	14.671
tdc_herg_karim	TDC	classification	Ensemble (stacking / averaging)	Ensemble (Weighted average (inverse OOF error))	test_ roc_ auc	0.905	ChemML MLP (Py Torch)	Deep tabular NN (ChemML MLP)	3.648
tdc_hia_hou	TDC	classification	CFA combinatorial fusion	CFA (Combinatorial Fusion)	test_ roc_ auc	0.987	Logistic Regression	Conventional ML	0.438
tdc_pampa_ncats	TDC	classification	Uni-Mol (3D pretrained)	Uni-Mol V1	test_ roc_ auc	0.763	Logistic Regression	Conventional ML	26.281
tdc_pgp_broccatelli	TDC	classification	Uni-Mol (3D pretrained)	Uni-Mol V2 (84m)	test_ roc_ auc	0.933	Logistic Regression	Conventional ML	6.966
tdc_skin_reaction	TDC	classification	Uni-Mol (3D pretrained)	Uni-Mol V2 (84m)	test_ roc_ auc	0.658	TabPFNClassifier	TabPFN (tabular foundation)	20.120
tdc_tox21	TDC	classification	Chemprop v2 GNN	Chemprop v2 (AttentiveFP, ensemble=3)	test_ roc_ auc	0.597	TabPFNClassifier	TabPFN (tabular foundation)	21.688
tdc_toxcast	TDC	classification	Conventional ML	HistGradientBoosting	test_ roc_ auc	0.716	TabPFNClassifier	TabPFN (tabular foundation)	8.338
chemml_cep_homo	ChemML	regression	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5-fold))	test_ rmse	0.093	TabPFNRegressor	TabPFN (tabular foundation)	9.478
chemml_organic_density	ChemML	regression	TabPFN (tabular foundation)	TabPFNRegressor	test_ rmse	0.005	TabPFNRegressor	TabPFN (tabular foundation)	0.000
esol_delaney	MoleculeNet	regression	Ensemble (stacking / averaging)	Ensemble (Weighted average (inverse OOF error))	test_ rmse	0.626	TabPFNRegressor	TabPFN (tabular foundation)	7.502
freesolv_sampler	MoleculeNet	regression	TabPFN (tabular foundation)	TabPFNRegressor	test_ rmse	0.933	TabPFNRegressor	TabPFN (tabular foundation)	0.000
lipophilicity	MoleculeNet	regression	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5-fold))	test_ rmse	0.548	Elastic NetCV	Conventional ML	24.523
poduam_pod_nc_std	PODUAM	regression	Conventional ML	Random forest	test_ rmse	0.720	TabPFNRegressor	TabPFN (tabular foundation)	3.415

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dataset	suite	task_ kind	family	model	analysis_ metric	analysis_ metric_value	cv_ selected_ model	cv_ selected_ family	cv_selected_gap_to_best
poduam_pod_ rd_std	PODUAM	regression	Ensemble (stacking / averaging)	Ensemble (Weighted average (inverse OOF error))	test_ rmse	0.570	TabPFNRegressor	TabPFN (tabular foundation)	3.252
polaris_adme_ fang_hppb_1	Polaris	regression	Chemprop v2 GNN	Chemprop v2 (D-MPNN, ensemble=3)	test_ rmse	0.444	Elastic NetCV	Conventional ML	23.269
polaris_adme_ fang_perm_1	Polaris	regression	Uni-Mol (3D pretrained)	Uni-Mol V1	test_ rmse	0.399	ChemML MLP (Py Torch)	Deep tabular NN (ChemML MLP)	15.413
polaris_adme_ fang_rclint_1	Polaris	regression	Chemprop v2 GNN	Chemprop v2 (D-MPNN, ensemble=3)	test_ rmse	0.517	TabPFNRegressor	TabPFN (tabular foundation)	2.507
polaris_adme_ fang_rppb_1	Polaris	regression	MapLight + GNN	MapLight + GNN (Cat Boost, Strict Parity)	test_ rmse	0.493	TabPFNRegressor	TabPFN (tabular foundation)	12.638
polaris_adme_ fang_solu_1	Polaris	regression	Uni-Mol (3D pretrained)	Uni-Mol V2 (84m)	test_ rmse	0.551	TabPFNRegressor	TabPFN (tabular foundation)	10.837
tdc_caco2_wang	TDC	regression	Ensemble (stacking / averaging)	Ensemble (Weighted average (inverse OOF error))	test_ rmse	0.345	TabPFNRegressor	TabPFN (tabular foundation)	11.441
tdc_clearance_ hepatocyte_az	TDC	regression	MapLight + GNN	MapLight + GNN (Cat Boost, Strict Parity)	test_ rmse	44.113	TabPFNRegressor	TabPFN (tabular foundation)	14.669
tdc_clearance_ microsome_az	TDC	regression	Uni-Mol (3D pretrained)	Uni-Mol V2 (84m)	test_ rmse	33.507	TabPFNRegressor	TabPFN (tabular foundation)	14.294
tdc_half_life_ obach	TDC	regression	Uni-Mol (3D pretrained)	Uni-Mol V1	test_ rmse	19.128	TabPFNRegressor	TabPFN (tabular foundation)	6.346
tdc_ hydrationfreeenergy_- freesolv	TDC	regression	Ensemble (stacking / averaging)	Ensemble (Weighted average (inverse OOF error))	test_ rmse	1.093	TabPFNRegressor	TabPFN (tabular foundation)	2.124
tdc_ld50_zhu	TDC	regression	TabPFN (tabular foundation)	TabPFNRegressor	test_ rmse	0.806	TabPFNRegressor	TabPFN (tabular foundation)	0.000
tdc_ lipophilicity_ astrazeneca	TDC	regression	Chemprop v2 GNN	Chemprop v2 (D-MPNN, ensemble=3)	test_ rmse	0.577	TabPFNRegressor	TabPFN (tabular foundation)	3.556
tdc_ppbr_az	TDC	regression	MapLight + GNN	MapLight + GNN (Cat Boost, Strict Parity)	test_ rmse	11.356	ChemML MLP (Py Torch)	Deep tabular NN (ChemML MLP)	27.931
tdc_solubility_ aqsolddb	TDC	regression	TabPFN (tabular foundation)	TabPFNRegressor	test_ rmse	0.961	TabPFNRegressor	TabPFN (tabular foundation)	0.000

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dataset	suite	task_ kind	family	model	analysis_ metric	analysis_ metric_ value	cv_ selected_ model	cv_ selected_ family	cv_selected_gap_to_best
tdc_vdss_ lombardo	TDC	regression	MapLight + GNN	MapLight + GNN (Cat Boost, Strict Parity)	test_ rmse	4.670	TabPFNRegress	TabPFN (tabular foundation)	6.060

Table S2: Staged component ablation: datasets whose best achievable primary metric improved when each pipeline stage was added, computed from the existing model results.

stage_order	stage	datasets_evaluated	datasets_improved_vs_previous	improvement_fraction
1	Conventional ML only	44	0	0.000
2	+ Map Light classic fea- tures	44	10	0.227
3	+ neu- ral/ deep back- ends	44	31	0.705
4	+ CFA fusion	44	3	0.068
5	Full pipeline incl. ensem- bles	44	13	0.295

Table S3: Feature-family selection summary across all datasets, sorted by per-feature enrichment relative to a uniform-selection baseline.

Feature family	Datasets selected	Available (sum)	Selected (sum)	% of selected	Enrichment vs uniform
rdkit	40	8,261	1,362	11.22	5.16
erg	38	12,552	1,146	9.44	2.86
avalon	43	44,384	3,603	29.69	2.54
maccs	38	5,962	224	1.85	1.18
maplight	41	30,070	854	7.04	0.89
fcfp6	43	42,500	998	8.22	0.73
atom_pair	41	39,651	830	6.84	0.66
rdk_path	40	45,036	896	7.38	0.62
topological_torsion	40	23,653	421	3.47	0.56
ecfp6	37	43,299	731	6.02	0.53
layered	40	43,642	674	5.55	0.48
morgan	38	40,827	398	3.28	0.31

Table S4: Model coverage: datasets attempted and datasets yielding a valid metric, per model.

model	datasets_attempted	datasets_valid
Uni-Mol V2 (310m)	8	0
Uni-Mol V2 (164m)	4	0
Uni-Mol V2 (84m)	44	17
ElasticNetCV	22	20
MapLight CatBoost (Strict Parity)	22	22
LogisticRegression	22	22
SVC	22	22
TabPFNRegressor	22	22
SVR	22	22
TabPFNClassifier	22	22
Voting Classifier (KNN, SVM)	22	22
Voting Regressor (KNN, SVM)	22	22
Tabular CNN	22	22
Chemprop v2 (D-MPNN + RDKit2D, ensemble=3)	44	38
CFA (Combinatorial Fusion)	38	38
Chemprop v2 (AttentiveFP, ensemble=3)	44	38
Chemprop v2 (CMPNN, ensemble=3)	44	42
Chemprop v2 (D-MPNN + Selected descriptors, ensemble=3)	44	42
Chemprop v2 (D-MPNN, ensemble=3)	44	42
Ensemble (OOF Stacking (RidgeCV, 5-fold))	44	44
ChemML MLP (PyTorch)	44	44
CatBoost	44	44
AdaBoost	44	44
ChemML MLP (TensorFlow)	44	44
HistGradientBoosting	44	44
Extra trees	44	44
Ensemble (Weighted average (inverse OOF error))	44	44
Random forest	44	44
MapLight + GNN (CatBoost, Strict Parity)	44	44
Uni-Mol V1	44	44
Tabular MLP	44	44
XGBoost	44	44

Table S5: Run-to-run comparison: the canonical NSF ACCESS Jetstream2 A100 benchmark against the earlier consumer-GPU (RTX 4060) run, analysed identically. "Same split" marks datasets where both runs used the identical held-out partition; the remainder were re-split to scaffold splits in the A100 run and are therefore not directly comparable. Change is relative and metric-direction aware (positive favours the A100 run).

Dataset	Task	Same split	A100 best model	A100 value	RTX 4060 best model	RTX 4060 value	Change (%)
tdc_ames	classification	True	TabPFNClassifier	0.876	XGBoost	0.875	0.166
tdc_bbb_martins	classification	True	Random forest	0.925	Random forest	0.932	-0.788
tdc_bioavailability_ma	classification	True	CatBoost	0.777	CatBoost	0.777	0.000
tdc_cyp1a2_veith	classification	True	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.973	Ensemble (Weighted average (inverse train RMSE))	0.970	0.301
tdc_cyp2c19_veith	classification	True	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.931	Ensemble (Weighted average (inverse train RMSE))	0.921	1.116
tdc_cyp2c9_substrate_carbonmangels	classification	True	Chemprop v2 (AttentiveFP, ensemble=3)	0.438	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.482	-8.990
tdc_cyp2c9_veith	classification	True	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.829	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.810	2.384
tdc_cyp2d6_substrate_carbonmangels	classification	True	Chemprop v2 (D-MPNN + RDKit2D, ensemble=3)	0.693	Ensemble (Weighted average (inverse train RMSE))	0.673	2.919
tdc_cyp2d6_veith	classification	True	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.761	XGBoost	0.722	5.317
tdc_cyp3a4_substrate_carbonmangels	classification	True	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.724	LogisticRegression	0.717	1.017
tdc_cyp3a4_veith	classification	True	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.902	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.890	1.439
tdc_dili	classification	True	Uni-Mol V1	0.922	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.915	0.760
tdc_herg	classification	True	Chemprop v2 (AttentiveFP, ensemble=3)	0.857	AdaBoost	0.861	-0.411
tdc_herg_karim	classification	True	Ensemble (Weighted average (inverse OOF error))	0.905	Ensemble (Weighted average (inverse train RMSE))	0.902	0.330
tdc_hia_hou	classification	True	CFA (Combinatorial Fusion)	0.987	CFA (Combinatorial Fusion)	0.990	-0.249
tdc_pgp_broccatelli	classification	True	Uni-Mol V2 (84m)	0.933	Ensemble (Weighted average (inverse train RMSE))	0.929	0.423
chemml_cep_homo	regression	True	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.093	TabPFNRegressor	0.086	-8.909
chemml_organic_density	regression	True	TabPFNRegressor	0.005	TabPFNRegressor	0.005	5.907
esol_delaney	regression	True	Ensemble (Weighted average (inverse OOF error))	0.626	Ensemble (Weighted average (inverse train RMSE))	0.592	-5.716
freesolv_sampl	regression	True	TabPFNRegressor	0.933	Chemprop v2 (AttentiveFP, ensemble=1)	1.080	13.562
lipophilicity	regression	True	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.548	Ensemble (Weighted average (inverse train RMSE))	0.582	5.848
poduam_pod_nc_std	regression	True	Random forest	0.720	Ensemble (Weighted average (inverse train RMSE))	0.699	-3.062

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Dataset	Task	Same split	A100 best model	A100 value	RTX 4060 best model	RTX 4060 value	Change (%)
poduam_pod_rd_std	regression	True	Ensemble (Weighted average (inverse OOF error))	0.570	XGBoost	0.551	-3.493
polaris_adme_fang_hppb_1	regression	True	Chemprop v2 (D-MPNN, ensemble=3)	0.444	MapLight + GNN (CatBoost, Strict Parity)	0.449	1.072
polaris_adme_fang_perm_1	regression	True	Uni-Mol V1	0.399	Uni-Mol V1	0.399	-0.040
polaris_adme_fang_rclint_1	regression	True	Chemprop v2 (D-MPNN, ensemble=3)	0.517	Uni-Mol V1	0.512	-0.968
polaris_adme_fang_rppb_1	regression	True	MapLight + GNN (CatBoost, Strict Parity)	0.493	MapLight + GNN (CatBoost, Strict Parity)	0.494	0.170
polaris_adme_fang_solu_1	regression	True	Uni-Mol V2 (84m)	0.551	Uni-Mol V1	0.575	4.202
tdc_caco2_wang	regression	True	Ensemble (Weighted average (inverse OOF error))	0.345	Ensemble (Weighted average (inverse train RMSE))	0.336	-2.388
tdc_clearance_hepatocyte_az	regression	True	MapLight + GNN (CatBoost, Strict Parity)	44.113	MapLight + GNN (CatBoost, Strict Parity)	43.976	-0.311
tdc_clearance_microsome_az	regression	True	Uni-Mol V2 (84m)	33.507	Uni-Mol V1	35.888	6.634
tdc_half_life_obach	regression	True	Uni-Mol V1	19.128	Uni-Mol V1	17.222	-11.067
tdc_ld50_zhu	regression	True	TabPFNRegressor	0.806	CFA (Combinatorial Fusion)	0.834	3.290
tdc_lipophilicity_astrazeneca	regression	True	Chemprop v2 (D-MPNN, ensemble=3)	0.577	Uni-Mol V1	0.587	1.698
tdc_ppbr_az	regression	True	MapLight + GNN (CatBoost, Strict Parity)	11.356	Ensemble (Weighted average (inverse train RMSE))	11.014	-3.108
tdc_solubility_aqsolddb	regression	True	TabPFNRegressor	0.961	Uni-Mol V1	0.989	2.807
tdc_vdss_lombardo	regression	True	MapLight + GNN (CatBoost, Strict Parity)	4.670	MapLight + GNN (CatBoost, Strict Parity)	4.668	-0.035
tdc_carcinogens_lagunin	classification	False	Chemprop v2 (AttentiveFP, ensemble=3)	0.929	AdaBoost	0.863	7.656
tdc_clintox	classification	False	CFA (Combinatorial Fusion)	0.974	CatBoost	0.949	2.644
tdc_pampa_ncats	classification	False	Uni-Mol V1	0.763	Ensemble (Weighted average (inverse train RMSE))	0.806	-5.258
tdc_skin_reaction	classification	False	Uni-Mol V2 (84m)	0.658	CFA (Combinatorial Fusion)	0.769	-14.470
tdc_tox21	classification	False	Chemprop v2 (AttentiveFP, ensemble=3)	0.597	XGBoost	0.812	-26.451
tdc_toxcast	classification	False	HistGradientBoosting	0.716	CatBoost	0.791	-9.562
tdc_hydrationfreeenergy_freesolv	regression	False	Ensemble (Weighted average (inverse OOF error))	1.093	CFA (Combinatorial Fusion)	0.556	-96.349

Table S6: Dataset catalog. "Ranking metric" is the metric used for cross-dataset model ranking (RMSE for regression; the dataset's designated primary classification metric otherwise); leaderboard comparisons in Table S7 use the leaderboard's own metric, which differs for several TDC regression tasks.

Dataset	Suite	Task	Molecules	Train	Test	Split	Target scale	Ranking metric	Best model	Best value	Leaderboard metric	QSARena (lb metric)	Est. rank	Best published	Best published model
tdc_ames	TDC	classification	7,278	5,821	1,457	predefined	raw	roc_auc	TabPFNClassifier	0.876	ROC_AUC	0.876	2	0.912	QW-MTL
tdc_bbb_martins	TDC	classification	2,030	1,624	406	predefined	raw	roc_auc	Random forest	0.925	ROC_AUC	0.925	2	0.941	MolGPS (3B)
tdc_bioavailability_ma	TDC	classification	640	512	128	predefined	raw	roc_auc	CatBoost	0.777	ROC_AUC	0.777	1	0.748	MaxQsaring
tdc_carcinogens_lagunin	TDC	classification	280	223	57	scaffold	raw	roc_auc	Chemprop v2 (AttentiveFP, ensemble=3)	0.929	ROC_AUC	0.929	1	0.848	FATE-Tox (MTL)
tdc_clintox	TDC	classification	1,478	1,180	298	scaffold	raw	roc_auc	CFA (Combinatorial Fusion)	0.974	ROC_AUC	0.974	3	0.996	PrismNet
tdc_cyp1a2_veith	TDC	classification	12,579	10,061	2,518	scaffold	raw	roc_auc	Ensemble (OOFS Stacking (RidgeCV, 5-fold))	0.973					
tdc_cyp2c19_veith	TDC	classification	12,665	10,131	2,534	scaffold	raw	roc_auc	Ensemble (OOFS Stacking (RidgeCV, 5-fold))	0.931					
tdc_cyp2c9_substrate_carbonmangels	TDC	classification	669	534	135	predefined	raw	auprc	Chemprop v2 (AttentiveFP, ensemble=3)	0.438	AUPRC	0.438	3	0.450	MaxQsaring
tdc_cyp2c9_veith	TDC	classification	12,092	9,673	2,419	predefined	raw	auprc	Ensemble (OOFS Stacking (RidgeCV, 5-fold))	0.829	AUPRC	0.829	3	0.877	MaxQsaring
tdc_cyp2d6_substrate_carbonmangels	TDC	classification	667	532	135	predefined	raw	auprc	Chemprop v2 (D-MPNN + RDKit2D, ensemble=3)	0.693	AUPRC	0.693	4	0.766	MaxQsaring
tdc_cyp2d6_veith	TDC	classification	13,130	10,504	2,626	predefined	raw	auprc	Ensemble (OOFS Stacking (RidgeCV, 5-fold))	0.761	AUPRC	0.761	2	0.811	MaxQsaring
tdc_cyp3a4_substrate_carbonmangels	TDC	classification	670	535	135	predefined	raw	auprc	Ensemble (OOFS Stacking (RidgeCV, 5-fold))	0.724	ROC_AUC	0.655	7	0.692	MoIE

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Dataset	Suite	Task	Molecules	Train	Test	Split	Target scale	Ranking metric	Best model	Best value	Leaderboard metric	QSARena (lb metric)	Est. rank	Best published	Best published model
tdc_cyp3a4_veith	TDC	classification	12,328	9,861	2,467	predefined	raw	aucprc	Ensemble (OOFS Stacking (RidgeCV, 5-fold))	0.902	AUPRC	0.902	2	0.923	MaxQsaring
tdc_dili	TDC	classification	475	379	96	predefined	raw	roc_auc	Uni-Mol V1	0.922	ROC_AUC	0.922	4	0.945	Meta-model (NIST)
tdc_herg	TDC	classification	655	523	132	predefined	raw	roc_auc	Chemprop v2 (AttentiveFP, ensemble=3)	0.857	ROC_AUC	0.857	3	0.880	MaxQsaring
tdc_herg_karim	TDC	classification	13,445	10,755	2,690	scaffold	raw	roc_auc	Ensemble (Weighted average (inverse OOF error))	0.905					
tdc_hia_hou	TDC	classification	578	461	117	predefined	raw	roc_auc	CFA (Combinatorial Fusion)	0.987	ROC_AUC	0.987	4	0.994	MiniMol (GINE)
tdc_pampa_ncats	TDC	classification	2,034	1,626	408	scaffold	raw	roc_auc	Uni-Mol V1	0.763					
tdc_pgp_broccatelli	TDC	classification	1,218	973	245	predefined	raw	roc_auc	Uni-Mol V2 (84m)	0.933	ROC_AUC	0.933	4	0.994	MiniMol (GINE)
tdc_skin_reaction	TDC	classification	404	289	115	scaffold	raw	roc_auc	Uni-Mol V2 (84m)	0.658	ROC_AUC	0.658	>10	0.741	FATE-Tox (MTL)
tdc_tox21	TDC	classification	7,258	5,797	1,461	scaffold	raw	roc_auc	Chemprop v2 (AttentiveFP, ensemble=3)	0.597	ROC_AUC	0.597	>10	0.867	PrismNet
tdc_toxcast	TDC	classification	1,731	1,357	374	scaffold	raw	roc_auc	HistGradient Boosting	0.716	ROC_AUC	0.716	1	0.714	PrismNet
chemml_cep_homo	ChemML	Regression	500	400	100	target_ -	raw	rmse	Ensemble (OOFS Stacking (RidgeCV, 5-fold))	0.093					
chemml_organic_density	ChemML	Regression	500	400	100	quartiles target_ -	log10	rmse	TabPFNRegressor	0.005					
esol_delaney	MoleculeNet	Regression	1,128	874	254	quartiles scaffold	raw	rmse	Ensemble (Weighted average (inverse OOF error))	0.626	RMSE	0.626	7	0.558	GCN
freesolv_sampl	MoleculeNet	Regression	642	513	129	random	raw	rmse	TabPFNRegressor	0.933					
lipophilicity	MoleculeNet	Regression	4,200	3,357	843	scaffold	raw	rmse	Ensemble (OOFS Stacking (RidgeCV, 5-fold))	0.548	RMSE	0.548	1	0.549	GCN
poduam_pod_nc_std	PODUAM	Regression	1,842	1,473	369	target_ -	raw	rmse	Random forest	0.720	RMSE	0.720	2	0.550	PODUAM BNN (PODnc)
						quartiles									

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Dataset	Suite	Task	Molecules	Train	Test	Split	Target scale	Ranking metric	Best model	Best value	Leaderboard metric	QSARena (lb metric)	Est. rank	Best published	Best published model
poduam_pod_rd_std	PODUM	regression	2,355	1,884	471	target_-raw	raw	rmse	Ensemble (Weighted average (inverse OOF error))	0.570	RMSE	0.570	2	0.410	PODUAM BNN (PODrd)
						quartiles									
polaris_adme_fang_hppb_1	Polaris	regression	1,808	1,446	362	predefined	raw	rmse	Chemprop v2 (D-MPNN, ensemble=3)	0.444	MSE	0.197	4	0.143	LateFusion-k5
polaris_adme_fang_perm_1	Polaris	regression	2,642	2,113	529	predefined	raw	rmse	Uni-Mol V1	0.399	MSE	0.159	4	0.113	1B_MPNN_MolGPS-ens_Large Mix
polaris_adme_fang_rclint_1	Polaris	regression	3,054	2,443	611	predefined	raw	rmse	Chemprop v2 (D-MPNN, ensemble=3)	0.517	MSE	0.268	6	0.204	LateFusion-k5
polaris_adme_fang_rppb_1	Polaris	regression	885	708	177	predefined	raw	rmse	MapLight + GNN (CatBoost, Strict Parity)	0.493	MSE	0.243	3	0.230	1B_MPNN_LargeMix-and-Phenomics
polaris_adme_fang_solu_1	Polaris	regression	2,173	1,738	435	predefined	raw	rmse	Uni-Mol V2 (84m)	0.551	MSE	0.304	7	0.222	1B_MPNN_MolGPS-ens_Large Mix
tdc_caco2_wang	TDC	regression	910	728	182	predefined	raw	rmse	Ensemble (Weighted average (inverse OOF error))	0.345	MAE	0.272	3	0.256	CaliciBoost
tdc_clearance_hepatocyte_az	TDC	regression	1,213	970	243	predefined	raw	rmse	MapLight + GNN (CatBoost, Strict Parity)	44.113	SPEARMAN	0.516	3	0.633	CFA
tdc_clearance_microsome_az	TDC	regression	1,102	881	221	predefined	raw	rmse	Uni-Mol V2 (84m)	33.507	SPEARMAN	0.678	1	0.652	MapLight + GNN
tdc_half_life_obach	TDC	regression	667	532	135	predefined	raw	rmse	Uni-Mol V1	19.128	SPEARMAN	0.597	2	0.649	CFA
tdc_hydrationfreeenergy_-freesolv	TDC	regression	642	490	152	scaffold	raw	rmse	Ensemble (Weighted average (inverse OOF error))	1.093	RMSE	1.093	7	0.654	PrismNet

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Dataset	Suite	Task	Molecules	Train	Test	Split	Target scale	Ranking met- ric	Best model	Best value	Leaderboard metric	QSARena (lb metric)	Est. rank	Best published	Best published model
tdc_ld50_ zhu	TDC	regression	7,385	5,907	1,478	predefined	raw	rmse	TabPFNRegressor	0.806	MAE	0.552	2	0.292	Base Boosting KyQVZ6b2 MiniMol
tdc_ lipophilicity_ astrazeneca	TDC	regression	4,200	3,360	840	predefined	raw	rmse	Chemprop v2 (D- MPNN, ensemble=3)	0.577	MAE	0.425	3	0.406	Gradient Boost
tdc_ppbr_az	TDC	regression	2,790	2,231	559	predefined	raw	rmse	MapLight + GNN (CatBoost, Strict Parity)	11.356	MAE	7.306	3	0.679	MiniMol
tdc_ solubility_ aqsoldb	TDC	regression	9,980	7,985	1,995	predefined	raw	rmse	TabPFNRegressor	0.961	MAE	0.686	2	0.557	MapLight + GNN
tdc_vdss_ lombardo	TDC	regression	1,130	904	226	predefined	raw	rmse	MapLight + GNN (CatBoost, Strict Parity)	4.670	SPEARMAN	0.680	5	0.942	

Table S7: Per-dataset leaderboard comparison, sorted by estimated rank. “References (n)” is the number of published values available for that dataset and metric; ranks from sparse reference sets are correspondingly uncertain. The **tdc_ppbr_az** top-1 reference (MAE 0.679) is inconsistent in scale with the rest of that dataset’s references (top-10 cutoff 7.914) and its top-1 gap should be disregarded.

Dataset	Metric	QSARena best model (test-selected)	QSARena value	Reference top-1	Reference top-10 cutoff	References (n)	Est. rank	CV-selected model	CV-selected value	CV-selected est. rank
lipophilicity	RMSE	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.548	0.549	0.610	27	1	Elastic NetCV	0.682	>10
tdc_bioavailability_ma	ROC - AUC	- CatBoost	0.777	0.748	0.640	8	1	TabPFNClassifier	0.732	2
tdc_carcinogens_lagunin	ROC - AUC	- Chemprop v2 (AttentiveFP, ensemble=3)	0.929	0.848	0.795	20	1	TabPFNClassifier	0.870	1
tdc_clearance_microsome_az	SPEARMA	- Mol V2 (84m)	0.678	0.652	0.599	17	1	TabPFNRegressor	0.437	>10
tdc_toxcast	ROC - AUC	- HistGradientBoosting	0.716	0.714	0.714	2	1	TabPFNClassifier	0.656	3
poduam_pod_nc_std	RMSE	- Random forest	0.720	0.550	0.730	2	2	TabPFNRegressor	0.745	3
poduam_pod_rd_std	RMSE	- Ensemble (Weighted average (inverse OOF error))	0.570	0.410	0.630	2	2	TabPFNRegressor	0.589	2
tdc_ames	ROC - AUC	- TabPFNClassifier	0.876	0.912	0.834	7	2	TabPFNClassifier	0.876	2
tdc_bbb_martins	ROC - AUC	- Random forest	0.925	0.941	0.903	7	2	TabPFNClassifier	0.909	4
tdc_cyp2d6_veith	AUPRC	- Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.761	0.811	0.464	6	2	TabPFNClassifier	0.457	7
tdc_cyp3a4_veith	AUPRC	- Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.902	0.923	0.750	6	2	TabPFNClassifier	0.740	7
tdc_half_life_obach	SPEARMA	- Mol V1	0.597	0.649	0.485	16	2	TabPFNRegressor	0.221	>10
tdc_ld50_zhu	MAE	- TabPFNRegressor	0.552	0.292	0.605	16	2	TabPFNRegressor	0.552	2
tdc_solubility_aqsoldb	MAE	- TabPFNRegressor	0.686	0.557	0.776	17	2	TabPFNRegressor	0.686	2
polaris_adme_fang_rppb_1	MSE	- MapLight + GNN (Cat Boost, Strict Parity)	0.243	0.230	0.634	10	3	TabPFNRegressor	0.308	5
tdc_caco2_wang	MAE	- MapLight CatBoost (Strict Parity)	0.272	0.256	0.288	20	3	TabPFNRegressor	0.302	>10
tdc_clearance_hepatocyte_az	SPEARMA	- MapLight + GNN (Cat Boost, Strict Parity)	0.516	0.633	0.440	16	3	TabPFNRegressor	0.186	>10
tdc_clintox	ROC - AUC	- CFA (Combinatorial Fusion)	0.974	0.996	0.889	12	3	TabPFNClassifier	0.923	3
tdc_cyp2c9_substrate_carbonmangels	AUPRC	- Chemprop v2 (AttentiveFP, ensemble=3)	0.438	0.450	0.360	6	3	SVC	0.301	7

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Dataset	Metric	QSARena best model (test-selected)	QSARena value	Reference top-1	Reference top-10 cutoff	References (n)	Est. rank	CV- selected model	CV-selected value	CV- selected est. rank
tdc_cyp2c9_veith	AUPRC	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.829	0.877	0.770	6	3	TabPFNClassifier	0.530	7
tdc_herg	ROC - AUC	Chemprop v2 (AttentiveFP, ensemble=3)	0.857	0.880	0.806	7	3	TabPFNClassifier	0.732	8
tdc_lipophilicity_ astrazeneca	MAE	Chemprop v2 (D-MPNN, ensemble=3)	0.425	0.406	0.515	17	3	TabPFNRegressor	0.439	4
tdc_ppbr_az	MAE	MapLight + GNN (Cat Boost, Strict Parity)	7.306	0.679	7.914	16	3	ChemML MLP (Py Torch)	9.741	>10
polaris_adme_ fang_hppb_1	MSE	Chemprop v2 (D-MPNN, ensemble=3)	0.197	0.143	0.303	10	4	Elastic NetCV	0.300	10
polaris_adme_ fang_perm_1	MSE	Uni-Mol V1	0.159	0.113	0.239	10	4	ChemML MLP (Py Torch)	0.212	9
tdc_cyp2d6_ substrate_ carbonmangels	AUPRC	Chemprop v2 (D-MPNN + RDKit2D, ensemble=3)	0.693	0.766	0.570	7	4	AdaBoost	0.620	7
tdc_dili	ROC - AUC	Uni-Mol V1	0.922	0.945	0.852	6	4	Logistic Regression	0.745	7
tdc_hia_hou	ROC - AUC	CFA (Combinatorial Fusion)	0.987	0.994	0.976	8	4	Logistic Regression	0.983	7
tdc_pgp_broccatelli	ROC - AUC	Uni-Mol V2 (84m)	0.933	0.994	0.911	8	4	Logistic Regression	0.868	9
tdc_vdss_lombardo	SPEARMAN	MapLight + GNN (Cat Boost, Strict Parity)	0.680	0.942	0.582	16	5	TabPFNRegressor	0.489	>10
polaris_adme_ fang_rclint_1	MSE	Chemprop v2 (D-MPNN, ensemble=3)	0.268	0.204	0.337	10	6	TabPFNRegressor	0.281	6
esol_delaney	RMSE	Ensemble (Weighted average (inverse OOF error))	0.626	0.558	0.743	30	7	TabPFNRegressor	0.673	9
polaris_adme_ fang_solu_1	MSE	Uni-Mol V2 (84m)	0.304	0.222	0.323	10	7	TabPFNRegressor	0.373	>10
tdc_cyp3a4_ substrate_ carbonmangels	ROC - AUC	Voting Classifier (KNN, SVM)	0.655	0.692	0.651	7	7	TabPFNClassifier	0.621	8
tdc_ hydrationfreeenergy_ freesolv	RMSE	Ensemble (Weighted average (inverse OOF error))	1.093	0.654	1.211	12	7	TabPFNRegressor	1.116	9
tdc_skin_reaction	ROC - AUC	Uni-Mol V2 (84m)	0.658	0.741	0.677	21	>10	TabPFNClassifier	0.525	>10
tdc_tox21	ROC - AUC	Chemprop v2 (AttentiveFP, ensemble=3)	0.597	0.867	0.840	12	>10	TabPFNClassifier	0.468	>10

Table S8: Applicability domain and calibration on the 22 official TDC ADMET Benchmark Group splits, for a fixed reference model (random forest, 300 trees, on Morgan radius-2 2048-bit fingerprints plus RDKit 2D descriptors) fitted to 80% of each train_val partition, with the remaining 20% used only for conformal calibration. Coverage is the fraction of test compounds inside the domain. The error ratio is the mean test error outside the domain divided by that inside it (MAE for regression, 0/1 error for classification); values above 1 mean the domain flag separates worse predictions. “Std.” is the Roy standardization approach on RDKit descriptors; “kNN” is the 5-nearest-neighbour Tanimoto distance thresholded at the 95th percentile of training distances; “Reliability” is the conformal-interval-width flag (regression) or the posterior-probability confidence flag with $|p_1 - p_0| \geq 0.5$ (classification). Conformal coverage is the empirical test coverage of 90% split-conformal intervals or prediction sets. “—” marks a ratio not reported because fewer than five test compounds fall on one side of the domain boundary. Results describe the reference model, not the benchmark’s selected model; single seed.

Dataset	Task	n test	Std. coverage	Std. ratio	kNN coverage	kNN ratio	Reliability ratio	Conformal coverage	ECE
caco2_wang	reg.	182	0.99	—	0.74	0.75	1.38	0.95	—
clearance_hepatocyte_az	reg.	243	0.94	1.09	0.88	1.26	1.75	0.94	—
clearance_microsome_az	reg.	221	0.99	—	0.97	1.32	2.08	0.93	—
half_life_obach	reg.	135	0.95	1.36	0.94	0.87	3.60	0.94	—
ld50_zhu	reg.	1478	0.95	1.16	0.82	1.24	2.11	0.82	—
lipophilicity_astazeneca	reg.	840	0.99	2.33	0.94	1.55	1.47	0.88	—
ppbr_az	reg.	559	0.99	1.35	0.91	1.05	3.01	0.93	—
solubility_aqsolddb	reg.	1995	0.95	1.45	0.86	1.15	1.59	0.89	—
vdss_lombardo	reg.	226	0.94	0.49	0.86	1.22	8.26	0.95	—
ames	cls.	1457	0.96	0.91	0.84	1.71	2.67	0.89	0.07
bbb_martins	cls.	406	0.96	0.00	0.78	1.19	9.30	0.91	0.06
bioavailability_ma	cls.	128	0.98	—	0.96	0.72	3.72	0.91	0.08
cyp2c9_substrate_carbonmangels	cls.	135	0.93	0.68	0.95	0.00	2.15	0.81	0.11
cyp2c9_veith	cls.	2419	0.96	0.53	0.88	0.62	5.21	0.91	0.05
cyp2d6_substrate_carbonmangels	cls.	135	0.96	0.65	0.98	—	4.82	0.89	0.06
cyp2d6_veith	cls.	2626	0.96	0.90	0.84	0.89	5.29	0.91	0.06
cyp3a4_substrate_carbonmangels	cls.	135	0.95	2.08	0.94	1.80	1.47	0.93	0.06
cyp3a4_veith	cls.	2467	0.98	0.91	0.94	0.93	4.88	0.89	0.07
dili	cls.	96	0.95	3.64	0.95	0.00	2.16	0.95	0.17
herg	cls.	132	0.99	—	0.92	1.10	4.71	0.86	0.10
hia_hou	cls.	117	0.97	—	0.82	4.57	6.13	0.91	0.09
pgp_broccatelli	cls.	245	0.94	0.75	0.84	1.21	5.04	0.88	0.06

Table S9: Dataset meta-feature catalog computed on each benchmark train/test partition (verified against the recorded split hashes). Diversity and similarity use Morgan radius-2, 2,048-bit fingerprints; SNN is the maximum Tanimoto similarity of a test molecule to any training molecule, and OOD fraction is the share of test molecules with SNN below 0.40.

Dataset	Task	Split	Metric	n train	n test	Positive fraction	Imbalance ratio	Target SD	Target skew	Scaffolds	Scaffolds / molecule	Singleton scaffold fraction	Internal diversity	Mean SNN	OOD fraction (SNN < 0.40)
tdc_ames	classification	patidefinedest_-	auc	5,821	1,457	0.533	1.142			2,387	0.410	0.809	0.916	0.517	0.251
			roc_-												
tdc_bbb_martins	classification	patidefinedest_-	auc	1,624	406	0.753	3.050			829	0.510	0.744	0.890	0.418	0.520
			roc_-												
tdc_bioavailability_ma	classification	patidefinedest_-	auc	512	128	0.771	3.376			363	0.709	0.837	0.899	0.435	0.492
			roc_-												
tdc_carcinogens_lagunin	classification	scaffolddest_-	auc	223	57	0.184	4.439			191	0.857	0.864	0.914	0.346	0.684
			roc_-												
tdc_clintox	classification	scaffolddest_-	auc	1,180	298	0.083	11.041			894	0.758	0.813	0.915	0.425	0.517
			roc_-												
tdc_cyp1a2_veith	classification	scaffolddest_-	auc	10,061	2,518	0.445	1.249			7,622	0.758	0.811	0.887	0.561	0.199
			roc_-												
tdc_cyp2c19_veith	classification	scaffolddest_-	auc	10,131	2,534	0.496	1.018			7,615	0.752	0.807	0.887	0.563	0.204
			roc_-												
			auc												

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Dataset	Task	Split	Metric	n train	n test	Positive fraction	Imbalance ratio	Target SD	Target skew	Scaffolds	Scaffolds / molecule	Singleton scaffold fraction	Internal diversity	Mean SNN	OOD fraction (SNN < 0.40)
tdc_cyp2c9_-	classification	patidefined_-	auprc	534	135	0.193	4.184			366	0.685	0.831	0.899	0.408	0.504
substrate_-															
carbonmangels															
tdc_cyp2c9_-	classification	patidefined_-	auprc	9,673	2,419	0.339	1.954			5,727	0.592	0.784	0.885	0.509	0.338
veith															
tdc_cyp2d6_-	classification	patidefined_-	auprc	532	135	0.278	2.595			373	0.701	0.845	0.898	0.426	0.526
substrate_-															
carbonmangels															
tdc_cyp2d6_-	classification	patidefined_-	auprc	10,504	2,626	0.197	4.072			6,643	0.632	0.779	0.884	0.487	0.364
veith															
tdc_cyp3a4_-	classification	patidefined_-	auprc	535	135	0.516	1.066			375	0.701	0.832	0.899	0.436	0.474
substrate_-															
carbonmangels															
tdc_cyp3a4_-	classification	patidefined_-	auprc	9,861	2,467	0.408	1.448			6,088	0.617	0.800	0.889	0.534	0.260
veith															
tdc_dili	classification	patidefined_-	auc	379	96	0.491	1.038			265	0.699	0.879	0.907	0.332	0.781
			roc_-												
tdc_herg	classification	patidefined_-	auc	523	132	0.677	2.095			341	0.652	0.792	0.895	0.484	0.379
			roc_-												
tdc_herg_karim	classification	patidefined_-	auc	10,755	2,690	0.492	1.031			5,884	0.547	0.646	0.878	0.672	0.084
			roc_-												
			auc												

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Dataset	Task	Split	Metric	n train	n test	Positive fraction	Imbalance ratio	Target SD	Target skew	Scaffolds	Scaffolds / molecule	Singleton scaffold fraction	Internal diversity	Mean SNN	OOD fraction (SNN < 0.40)
tdc_hia_hou	classification	patidefinedest_-	auc roc_-	461	117	0.889	8.039			317	0.688	0.868	0.894	0.371	0.598
tdc_pampa_ncats	classification	scaffolddest_-	auc roc_-	1,626	408	0.846	5.504			1,355	0.833	0.862	0.871	0.642	0.064
tdc_pgp_broccatelli	classification	patidefinedest_-	auc roc_-	973	245	0.541	1.177			546	0.561	0.764	0.874	0.490	0.400
tdc_skin_reaction	classification	scaffolddest_-	auc roc_-	289	115	0.661	1.949			247	0.855	0.895	0.918	0.312	0.904
tdc_tox21	classification	scaffolddest_-	auc roc_-	5,797	1,461	0.049	19.412			3,853	0.665	0.872	0.925	0.467	0.305
tdc_toxcast	classification	scaffolddest_-	auc roc_-	1,357	374	0.262	2.812			1,002	0.738	0.889	0.926	0.439	0.422
chemml_cep_homo	regression	target_test_-	auc rmse	400	100			0.687	0.029	400	1	1	0.844	0.492	0.220
chemml_organic_density	regression	target_test_-	auc rmse	400	100			0.031	-0.093	399	0.998	0.997	0.841	0.463	0.250
esol_delaney	regression	scaffolddest_-	auc rmse	874	254			2.247	-0.435	583	0.667	0.873	0.930	0.437	0.417
freesolv_sampling	regression	randomtest_-	rmse	513	129			3.677	-0.753	308	0.600	0.932	0.923	0.570	0.163

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Dataset	Task	Split	Metric	n train	n test	Positive fraction	Imbalance ratio	Target SD	Target skew	Scaffolds	Scaffolds / molecule	Singleton scaffold fraction	Internal diversity	Mean SNN	OOD fraction (SNN < 0.40)
lipophilicity	regression	scaffold	test_rmse	3,357	843			1.190	-0.557	2,355	0.702	0.791	0.882	0.584	0.152
poduam_nc_std	regression	scaffold	test_rmse	1,473	369			1.109	-1.529	945	0.642	0.899	0.922	0.521	0.322
poduam_nc_std	regression	quartiles	test_rmse	1,884	471			0.865	-1.743	1,237	0.657	0.905	0.917	0.549	0.268
polaris_adme_fang_hppb_1	regression	undefined	test_rmse	1,446	362			0.719	-0.337	1,003	0.694	0.827	0.878	0.583	0.331
polaris_adme_fang_perm_1	regression	undefined	test_rmse	2,113	529			0.682	0.746	1,885	0.892	0.926	0.875	0.443	0.465
polaris_adme_fang_rclint_1	regression	undefined	test_rmse	2,443	611			0.751	-0.134	2,146	0.878	0.922	0.876	0.444	0.435
polaris_adme_fang_rppb_1	regression	undefined	test_rmse	708	177			0.751	-0.435	569	0.804	0.872	0.877	0.511	0.429
polaris_adme_fang_solu_1	regression	undefined	test_rmse	1,738	435			0.668	-1.657	1,531	0.881	0.925	0.874	0.446	0.492
tdc_caco2_wang	regression	undefined	test_rmse	728	182			0.797	-0.715	402	0.552	0.694	0.890	0.459	0.473
tdc_clearance	regression	undefined	test_rmse	970	243			50.287	1.216	601	0.620	0.687	0.877	0.425	0.580
hepatocyte_az_tdc_clearance	regression	undefined	test_rmse	881	221			45.128	1.548	587	0.666	0.818	0.871	0.520	0.321
microsome_az															

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Dataset	Task	Split	Metric	n train	n test	Positive fraction	Imbalance ratio	Target SD	Target skew	Scaffolds	Scaffolds / molecule	Singleton scaffold fraction	Internal diversity	Mean SNN	OOD fraction (SNN < 0.40)
tdc_half_life_obach	regression	undefined	rmse	532	135			90.920	9.338	355	0.667	0.820	0.899	0.413	0.533
tdc_hydrationfreeenergy_freesolv	regression	scaffold	rmse	490	152			4.014	-1.427	381	0.778	0.934	0.927	0.384	0.625
tdc_ld50_zhu	regression	undefined	rmse	5,907	1,478			0.921	0.928	3,136	0.531	0.894	0.917	0.448	0.438
tdc_lipophilicity	regression	undefined	rmse	3,360	840			1.204	-0.582	1,932	0.575	0.769	0.879	0.575	0.219
astrazeneca_tdc_ppbr_az	regression	undefined	rmse	2,231	559			18.785	-1.962	946	0.424	0.498	0.879	0.520	0.349
tdc_solubility_aqsolddb	regression	undefined	rmse	7,985	1,995			2.369	-0.587	4,265	0.534	0.900	0.922	0.475	0.327
tdc_vdss_lombardo	regression	undefined	rmse	904	226			24.829	24.663	613	0.678	0.822	0.900	0.384	0.619

Table S10: Exploratory family $\times$ meta-feature associations: Spearman correlation between each meta-feature and the relative gap of the family’s best model to the per-dataset best, with a dataset-bootstrap 95% CI (10,000 resamples), a two-sided label-permutation p-value and a Benjamini–Hochberg q-value over the whole grid. Positive ρ means the family falls further behind as the meta-feature increases.

Model family	Meta-feature	Datasets	Spearman rho	95% CI	Permutation p	BH q
Conventional ML	log10_n_train	44	-0.024	-0.32 to 0.31	0.875	0.942
Conventional ML	internal_diversity	44	-0.316	-0.60 to 0.00	0.036	0.208
Conventional ML	mean_snn	44	0.226	-0.10 to 0.53	0.144	0.434
Conventional ML	ood_fraction	44	-0.332	-0.61 to 0.00	0.028	0.208
Conventional ML	scaffolds_per_molecule	44	0.331	0.03 to 0.57	0.031	0.208
Conventional ML	label_asymmetry	44	-0.191	-0.45 to 0.11	0.215	0.465
Ensemble (stacking / averaging)	log10_n_train	44	-0.401	-0.66 to -0.09	0.005	0.208
Ensemble (stacking / averaging)	internal_diversity	44	0.047	-0.28 to 0.35	0.756	0.879
Ensemble (stacking / averaging)	mean_snn	44	-0.314	-0.58 to -0.00	0.039	0.208
Ensemble (stacking / averaging)	ood_fraction	44	0.325	-0.00 to 0.61	0.032	0.208
Ensemble (stacking / averaging)	scaffolds_per_molecule	44	0.106	-0.17 to 0.37	0.491	0.732
Ensemble (stacking / averaging)	label_asymmetry	44	0.325	0.00 to 0.60	0.032	0.208
CFA combinatorial fusion	log10_n_train	38	0.072	-0.26 to 0.40	0.653	0.804
CFA combinatorial fusion	internal_diversity	38	-0.113	-0.45 to 0.23	0.497	0.732
CFA combinatorial fusion	mean_snn	38	0.305	-0.03 to 0.59	0.059	0.259
CFA combinatorial fusion	ood_fraction	38	-0.320	-0.63 to 0.04	0.049	0.236
CFA combinatorial fusion	scaffolds_per_molecule	38	0.140	-0.18 to 0.45	0.398	0.682
CFA combinatorial fusion	label_asymmetry	38	0.162	-0.17 to 0.46	0.330	0.633
Uni-Mol (3D pretrained)	log10_n_train	44	-0.144	-0.45 to 0.17	0.353	0.652
Uni-Mol (3D pretrained)	internal_diversity	44	0.243	-0.11 to 0.57	0.114	0.400
Uni-Mol (3D pretrained)	mean_snn	44	0.006	-0.31 to 0.31	0.971	0.971
Uni-Mol (3D pretrained)	ood_fraction	44	-0.085	-0.40 to 0.25	0.584	0.757
Uni-Mol (3D pretrained)	scaffolds_per_molecule	44	0.016	-0.32 to 0.36	0.919	0.942
Uni-Mol (3D pretrained)	label_asymmetry	44	0.042	-0.28 to 0.37	0.781	0.879
TabPFN (tabular foundation)	log10_n_train	44	0.243	-0.13 to 0.55	0.117	0.400
TabPFN (tabular foundation)	internal_diversity	44	-0.283	-0.55 to 0.02	0.066	0.262
TabPFN (tabular foundation)	mean_snn	44	0.192	-0.11 to 0.47	0.223	0.465
TabPFN (tabular foundation)	ood_fraction	44	-0.139	-0.45 to 0.20	0.371	0.660
TabPFN (tabular foundation)	scaffolds_per_molecule	44	0.106	-0.23 to 0.43	0.487	0.732
TabPFN (tabular foundation)	label_asymmetry	44	-0.211	-0.49 to 0.09	0.163	0.435
Chemprop v2 GNN	log10_n_train	42	-0.080	-0.37 to 0.23	0.613	0.774
Chemprop v2 GNN	internal_diversity	42	0.041	-0.28 to 0.37	0.787	0.879
Chemprop v2 GNN	mean_snn	42	-0.016	-0.33 to 0.30	0.922	0.942
Chemprop v2 GNN	ood_fraction	42	-0.020	-0.34 to 0.29	0.898	0.942

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Model family	Meta-feature	Datasets	Spearman rho	95% CI	Permutation p	BH q
Chemprop v2 GNN	scaffolds_per_molecule	42	0.043	-0.25 to 0.34	0.785	0.879
Chemprop v2 GNN	label_asymmetry	42	0.333	-0.00 to 0.61	0.027	0.208
MapLight + GNN	log10_n_train	44	-0.213	-0.46 to 0.08	0.161	0.435
MapLight + GNN	internal_diversity	44	0.154	-0.17 to 0.45	0.318	0.633
MapLight + GNN	mean_snn	44	-0.102	-0.40 to 0.21	0.506	0.732
MapLight + GNN	ood_fraction	44	-0.096	-0.39 to 0.22	0.535	0.732
MapLight + GNN	scaffolds_per_molecule	44	0.222	-0.07 to 0.49	0.145	0.434
MapLight + GNN	label_asymmetry	44	-0.092	-0.39 to 0.24	0.549	0.732
Deep tabular NN (ChemML MLP)	log10_n_train	44	-0.376	-0.62 to -0.08	0.013	0.208
Deep tabular NN (ChemML MLP)	internal_diversity	44	-0.194	-0.46 to 0.10	0.209	0.465
Deep tabular NN (ChemML MLP)	mean_snn	44	-0.098	-0.40 to 0.22	0.530	0.732
Deep tabular NN (ChemML MLP)	ood_fraction	44	0.096	-0.24 to 0.41	0.540	0.732
Deep tabular NN (ChemML MLP)	scaffolds_per_molecule	44	0.198	-0.13 to 0.49	0.193	0.465
Deep tabular NN (ChemML MLP)	label_asymmetry	44	0.200	-0.12 to 0.49	0.198	0.465

Table S11: Family selector (v2), leave-one-dataset-out regret: the relative gap (%) of the recommended family to the per-dataset best. Rows are pre-specified variants (model | feature blocks: F0 size, similarity, label asymmetry and task; F1 chemistry; F2 label landscape; F3 training-set CV landmarks); “sbs” always picks the family with the best record on the other datasets. The nested row chooses the variant inside an inner leave-one-dataset-out loop and is the only unbiased estimate; single variants are exploratory.

Selector (model feature blocks)	Mean regret (%)	Median regret (%)	Pick within 5% of best (%)	SBS-to-oracle gap closed
sbs	5.182	2.624	61.364	0
knn F0	7.591	2.201	61.364	-0.465
ridge F0	5.602	2.618	61.364	-0.081
rf F0	4.392	2.822	70.455	0.152
knn F0+F1	5.338	1.586	75	-0.030
ridge F0+F1	5.794	3.034	65.909	-0.118
rf F0+F1	5.695	1.991	72.727	-0.099
knn F0+F2	5.101	1.870	65.909	0.016
ridge F0+F2	4.949	2.035	68.182	0.045
rf F0+F2	5.407	3.580	63.636	-0.043
knn F0+F3	6.730	2.624	63.636	-0.299
ridge F0+F3	4.204	2.618	65.909	0.189
rf F0+F3	4.705	3.638	59.091	0.092
knn F0-F3	5.214	1.836	70.455	-0.006
ridge F0-F3	6.067	2.618	63.636	-0.171
rf F0-F3	5.267	3.577	59.091	-0.016
tree_cls F0	5.907	2.594	70.455	-0.140
nested (headline)	6.495	3.037	63.636	-0.253